Windows of the Hermite–Padé Solid: The Number-Wall Window Theorem in Simplicial Coordinates

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Abstract

A number wall is the array of Toeplitz determinants of a sequence. Its zeros come in square *windows*, every window has a closed frame of nonzero entries, and the frame entries form geometric sequences (Conway–Guy; Lunnon). We place the number wall inside a family of solids, one for each $N \geq 1$. For $N+1$ power series $f_0 = 1, f_1, \dots, f_N$, the **Hermite–Padé solid** is the function τ on $\mathbb{N}^{N+1}$ given by the Hermite–Padé (Mahler) determinants. Its level sets $|n| = \sigma$ are the dilated N-simplices of the simplicial frame, so the solid is a stack of simplex grids. For $N = 1$ it is the (one-sided) number wall. We prove:

1. **Hirota–Miwa.** τ satisfies the discrete KP equation $\tau_i\tau_{jk} - \tau_j\tau_{ik} + \tau_k\tau_{ij} = 0$ on every triple of directions, with no sign gauge (Doliwa–Siemaszko proved the equivalent statement for the approximants). The number wall's cross rule $\mathrm{NS} = \mathrm{C}^2 - \mathrm{EW}$ is the same equation for three column families (unit vectors, right shifts, left shifts) after one translation is quotiented out.
2. **Shape.** The zero set is the union of **elementary windows**, one for each irreducible deficient vector g. Each is the set of lattice points of the bipyramid $d + \mathrm{conv}\{0, se_0, \dots, se_N, \frac{s}{\kappa}\mathbf{1}_S\}$, where S is the support of g and $\kappa = |S| - 1$. In the one-gauge chart a window is a **breathing simplex**: its order slices are simplices homothetic to the ambient one. They grow at rate 1 to side s and then contract at rate κ toward the barycentre of the face S. Equivalently, the window is $\{|x^\natural| + \kappa\,\mathfrak{m}_S(x) \leq s\}$, with $x^\natural$ the standard form and $\mathfrak{m}_S$ the massaction (Paper 01e). The square is the case $N = 1$, where the two rates coincide.
3. **Depth.** Inside a window the corank of the determinant is $1 + \min(\mathfrak{m}_S(x),\, s - |x^\natural| - \kappa\mathfrak{m}_S(x))$, a tent over the bipyramid. In general it is half the ℓ^1 norm of the shifted-degree vector of a reduced order basis.
4. **Separation becomes a rank bound.** At most N independent windows meet any unit cube $n + \{0,1\}^{N+1}$. For $N = 1$ this gives the classical statement: windows are disjoint squares, separated in all eight directions, and the depth formula holds everywhere. For $N \geq 2$ the bound is sharp. Windows touch and overlap, as exact examples over $\mathbb{Q}$ show.
5. **The frame does not generalize naively.** The geometric facet law fails for $N = 2$; we state the frame theorem for $N \geq 2$ as open.

The shapes of elementary and combined singular blocks in two-dimensional cuts, and the failure of non-intersection beyond Padé, are in Beckermann (1990) and Beckermann–Labahn (1992). Our contributions are the full-dimensional polytope and its gauge reading, the depth tent, the unit-cube rank bound, the cross rule as a Hirota reduction, and the frame obstruction. A certificate (10/10) checks every statement, exactly over $\mathbb{Q}$ and over $\mathbb{F}_p$.

Keywords: number wall, Toeplitz determinant, Padé table, Hermite–Padé approximation, perfect system, discrete KP equation, Hirota–Miwa equation, octahedron recurrence, Dodgson condensation, simplicial coordinates, standard form, massaction, lattice polytope.

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1. Introduction

1.1 Number walls

Let $c = (c_k)_{k\in\mathbb{Z}}$ be a sequence over a field K. Its number wall is the array

$$S(m,q) = \det\left[c_{q+i-j}\right]_{0\le i,j<m}, \qquad S(0,q) = 1,$$

of Toeplitz determinants [Conway–Guy; Lunnon 2001]. Three facts organize it. **(Cross rule)** $S(m+2,q+1)S(m,q+1) = S(m+1,q+1)^2 - S(m+1,q+2)S(m+1,q)$, a case of the Desnanot–Jacobi identity. **(Windows)** The zeros form squares with sides parallel to the axes. **(Frame)** Each window is surrounded by a closed ring of nonzero entries whose sides are geometric sequences, with one multiplicative relation among the four ratios. Lunnon used these facts to compute walls across windows and to read linear complexity off them. More recently, number walls over finite fields have become a tool in Diophantine approximation: they gave the counterexample to the t-adic Littlewood conjecture over $\mathbb{F}_3$ [Adiceam–Nesharim–Lunnon], its extensions [Garrett–Robertson; Robertson; Adiceam–Badziahin], and escape-of-mass results [Nesharim–Shapira–Soffer Aranov; Robertson–Soffer Aranov].

All of that work is two-dimensional. The question behind this paper is what the three facts become in more dimensions, and what the answer looks like in simplicial coordinates.

1.2 The answer in one paragraph

The number wall is the $N = 1$ member of a family. For series $f_0 = 1, f_1, \dots, f_N$, consider the Hermite–Padé problem: find polynomials a_i with $\deg a_i < n_i$ such that $\sum_i a_i f_i$ vanishes to order $|n|$. Its determinant $\tau(n)$ is a function on $\mathbb{N}^{N+1}$, the **Hermite–Padé solid**. The slice $|n| = \sigma$ is the σ-dilated simplex grid in barycentric coordinates, which is the one-gauge chart of the simplicial frame. The cross rule becomes the Hirota–Miwa equation. The square window becomes a breathing simplex: a bipyramid made of two simplices glued along an order-s facet. The frame's closedness becomes a rank bound, and the geometric frame law does not survive in the naive form.

1.3 Results

1. **Theorem 3.2 (Hirota–Miwa)** and **Proposition 3.3 (the cross rule is Hirota–Miwa).**
2. **Theorem 4.3 (union of elementary windows)** and **Theorem 4.4 (shape: bipyramid, slices, gauge form, size).**

3. **Theorem 4.5 (depth tent)** and **Proposition 4.6 (general deficiency).**
4. **Theorem 5.1 (local rank bound)**, **Corollary 5.3 (the classical window theorem,** $N = 1$**)** and **Proposition 5.4 (touching and overlap,** $N \geq 2$**).**
5. **Observation 6.1 (the frame law fails for** $N = 2$**)** and **Problem 6.2.**

1.4 What is classical and what is new

Classical. The block structure of the Padé table [Padé; Gragg 1972; Baker–Graves-Morris]; the multidimensional Padé table and its normality [Jager 1964; Mahler 1968, "perfect systems"]; the minimal-solution analysis of the singular M-Padé table [Beckermann 1990]. Beckermann and Labahn [BL 1992] work in a two-dimensional cut of the Hermite–Padé table. They show that the *elementary* singular block of an approximant P (the positions where P itself is an entry) is a triangle, and the *combined* block (where some $z^j P$ is an entry) is a union of triangles. Maximal combined blocks come from irreducible approximants. Non-intersection holds for Padé tables and fails in general. Their Thm. 3.1 is the dimension formula behind our Proposition 4.6. The Hirota (discrete KP) structure of Hermite–Padé approximation is due to [Doliwa–Siemaszko 2023]. Dodgson condensation is the octahedron recurrence [Robbins–Rumsey; Speyer].

New here, weighed honestly. **(a)** The elementary window in full dimension as an explicit lattice polytope: its V- and H-descriptions, the two-rate slice law, volume and lattice count, and the gauge form $|x^\natural| + \kappa \mathfrak{m}_S \leq s$ that ties it to the standard form and massaction of Paper 01e. The defining inequality follows from the definitions, as Beckermann–Labahn note for their cut; what we add is the geometry. **(b)** The depth tent, which makes the dimension formula explicit inside a window. **(c)** The unit-cube rank bound (Theorem 5.1). It recovers the separation half of the classical window theorem in one stroke and quantifies its failure for $N \geq 2$; we did not find it stated. **(d)** The derivation of the Toeplitz cross rule as a translation reduction of a three-family Hirota–Miwa system, with explicit signs. **(e)** The frame obstruction (Observation 6.1), stated as a negative finding with data.

1.5 Notation

K is a field. $|n| = \sum_i n_i$, and $n \leq n'$ is the componentwise order. e_i is a unit vector, $\mathbf{1}_S = \sum_{i \in S} e_i$, $\mathbf{1} = \mathbf{1}_{\{0,\dots,N\}}$. For $a \in K[z]^{N+1}$: $a \cdot f = \sum_i a_i f_i$, ord is the z-adic order, and $\deg 0 = -\infty$. $\Delta_s = \{x \in \mathbb{R}^{N+1}_{\geq 0} : |x| \leq s\}$ is the corner simplex, and $\Delta^N = \{y \geq 0 : |y| = 1\}$. The **massaction** on a face S is $\mathfrak{m}_S(x) = \min_{i \in S} x_i$, and the face-$S$ standard form is $x^\natural = x - \mathfrak{m}_S(x)\mathbf{1}_S$ (Paper 01e, notation; for S the full index set these are the genesis massaction and standard form).

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2. The Hermite–Padé solid

Definition 2.1. Let $f = (f_0, \dots, f_N) \in K[[z]]^{N+1}$ with $f_0(0) \neq 0$. For $n \in \mathbb{N}^{N+1}$ let $\mathcal{P}_n = \{a \in K[z]^{N+1} : \deg a_i < n_i\}$ (dimension $|n|$) and let $\Phi_n : \mathcal{P}_n \to K^{|n|}$ send a to the first $|n|$ coefficients of $a \cdot f$. Write $M(n)$ for the matrix of Φ_n. Its rows are $k = 0, \dots, |n|-1$, its columns are the pairs (i, j) with $j < n_i$ in block order (i first, then j), and its $(k, (i, j))$ entry is $[z^k]\, z^j f_i$. The **Hermite–Padé solid** of f is

$$\tau(n) = \det M(n), \qquad \tau(0) = 1.$$

Lemma 2.2 (kernel). $\ker \Phi_n = K(n) := \{a : \deg a_i < n_i,\ \operatorname{ord}(a \cdot f) \geq |n|\}$. *Hence* $\tau(n) = 0$ *if and only if* $K(n) \neq 0$*, and the deficiency* $\delta(n) := |n| - \operatorname{rank} M(n) = \dim K(n)$. □

Example 2.3 (the number wall is the case $N = 1$**).** Take $f_0 = 1$ and $f_1 = \sum_{k \geq 0} c_k z^k$, and set $c_k = 0$ for $k < 0$. The unit columns $z^j \cdot 1$, $j < n_0$, clear the first n_0 rows, so $M(n)$ is block triangular and

$$\tau(n_0, n_1) = \det \left[c_{n_0 + i - j}\right]_{0 \leq i, j < n_1} = S(n_1, n_0).$$

The $N = 1$ solid is the number wall of $(\dots, 0, 0, c_0, c_1, \dots)$, which is the classical Padé C-table up to indexing. Two-sided walls are treated in Proposition 3.3.

The one-gauge chart. The level set $\Sigma_\sigma = \{n \in \mathbb{N}^{N+1} : |n| = \sigma\}$ is the σ-dilate of the standard simplex grid, written in barycentric (one-gauge) coordinates. It is a segment for $N = 1$, the triangular grid for $N = 2$ and the tetrahedral grid for $N = 3$. The solid is the cone of these grids, with the order σ as height. Adjoining $n_{N+1} = -|n|$ places n in the root lattice $Q(A_{N+1}) = \{x \in \mathbb{Z}^{N+2} : \sum x_i = 0\}$. The order is the gauge partner of the degrees, and the steps e_i become the roots $e_i - e_{N+1}$; compare Doliwa's A-type formulation of the Hirota system [Doliwa 2010]. **Mahler's perfect systems** [Mahler 1968], those with $\tau(n) \neq 0$ for all n, are exactly the windowless solids.

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3. The Hirota–Miwa equation

Consider any matrix with rows indexed by $k \geq 0$. For an ordered list of columns C with $|C| = r$, let $D(C)$ be the determinant of the top r rows of C (a *top-justified* minor). We write $D(X; u)$ and $D(X; u, v)$ for X followed by the appended columns.

Lemma 3.1 (three-term identity for top-justified minors). *For any list X and columns a, b, c,*

$$D(X; a)\, D(X; b, c) - D(X; b)\, D(X; a, c) + D(X; c)\, D(X; a, b) = 0.$$

Proof. Both sides are polynomials in the entries, so we may assume the top $r \times r$ block X_0 of X is invertible. Subtracting from a, b, c the combinations of the columns of X that clear their top r entries changes none of the six minors. After that, $D(X; u) = \det X_0 \cdot u_r$ and $D(X; u, v) = \det X_0 \cdot (u_r v_{r+1} - u_{r+1} v_r)$, because the matrices are block lower triangular. The identity becomes $(\det X_0)^2$ times the expansion of

$$\det \begin{pmatrix} a_r & b_r & c_r \\ a_r & b_r & c_r \\ a_{r+1} & b_{r+1} & c_{r+1} \end{pmatrix} = 0. \qquad \square$$

Theorem 3.2 (Hirota–Miwa). *For all $n \in \mathbb{N}^{N+1}$ and $0 \leq i < j < k \leq N$,*

$$\tau(n + e_i)\, \tau(n + e_j + e_k) - \tau(n + e_j)\, \tau(n + e_i + e_k) + \tau(n + e_k)\, \tau(n + e_i + e_j) = 0.$$

Proof. Apply Lemma 3.1 to the infinite matrix whose columns are the coefficient vectors of $z^j f_i$. Take X to be the columns of $M(n)$ in block order and a, b, c the next columns $(i, n_i), (j, n_j), (k, n_k)$ of their blocks. Let $N_{>l} = \sum_{l' > l} n_{l'}$. Moving an appended column (l, n_l) to the end of block l passes $N_{>l}$ columns, so $D(X; a) = (-1)^{N_{>i}} \tau(n + e_i)$. For two appended columns $(l, n_l), (l', n_{l'})$ with $l < l'$, first move the second (past $N_{>l'}$ columns), then the first (past $N_{>l} + 1$). This gives $D = (-1)^{N_{>l} + N_{>l'} + 1} \tau(n + e_l + e_{l'})$. All three products in Lemma 3.1 then carry the same sign $(-1)^{N_{>i} + N_{>j} + N_{>k} + 1}$, which cancels. $\square$

This is the discrete KP (Hirota–Miwa) equation, equivalently the octahedron recurrence, on every three-dimensional sublattice. [Doliwa–Siemaszko 2023] showed that Hermite–Padé forms solve the Hirota system and its adjoint linear problem. Theorem 3.2 is the corresponding statement for the τ-function, with the sign conventions fixed. Certified over $\mathbb{Q}$ and $\mathbb{F}_p$ (W1, W2).

Proposition 3.3 (the cross rule is Hirota–Miwa). *Let $(c_k)_{k \in \mathbb{Z}}$ be bi-infinite and $v_j = (c_{k-j})_{k \geq 0}$. Form the three-family solid $\tau_3(n_0, n_1, n_2)$: the top-justified minor on the unit columns $e_0, \dots, e_{n_0 - 1}$, then $v_0, \dots, v_{n_1 - 1}$, then $v_{-1}, \dots, v_{-n_2}$, in that order. Then*

$$\tau_3(n) = (-1)^{n_1 n_2 + \binom{n_2}{2}}\, S(n_1 + n_2,\ n_0 + n_2),$$

τ_3 satisfies the Hirota–Miwa relation of Theorem 3.2, and under this substitution that relation is exactly the cross rule $S(m+2,q+1)S(m,q+1) = S(m+1,q+1)^2 - S(m+1,q+2)S(m+1,q)$.

Proof. The unit columns clear the first n_0 rows. Reordering the shift columns as $v_{-n_2}, \dots, v_{n_1-1}$ costs $\binom{n_2}{2}+n_1n_2$ transpositions, and leaves the Toeplitz matrix of size $m = n_1+n_2$ at offset $q = n_0+n_2$. The Hirota relation holds by Lemma 3.1 with the same bookkeeping as in Theorem 3.2. Substitute $\tau_3(n+v) = \pm S(\cdot,\cdot)$ for the six neighbours. With $A = n_1n_2 + \binom{n_2}{2}$ the signs are $(-1)^A$, $(-1)^{A+n_1+1}$, $(-1)^{A+n_2}$, $(-1)^{A+n_1+n_2}$, $(-1)^{A+n_1+n_2}$ and $(-1)^{A+n_2}$, for e_0, e_1+e_2, e_1, e_0+e_2, e_2 and e_0+e_1 respectively. Multiplying the relation by $(-1)^{n_1+1}$ gives the cross rule. The square $S(m+1,q+1)^2$ appears because $n+e_2$ and $n+e_0+e_1$ have the same (m,q). □

So the number wall is the three-family Hirota–Miwa solid modulo the translation $e_2 - e_0 - e_1$. This makes precise the folklore that Dodgson condensation of Toeplitz determinants is a reduction of the octahedron recurrence. Certified exactly (W10).

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4. The window theorem

Definition 4.1 (generators). A vector $g \in K[z]^{N+1} \setminus 0$ is *primitive* if its components have no common polynomial factor (in particular $z \nmid g$). Its data are $d(g)_i = \deg g_i + 1$ (with 0 for $g_i = 0$), $\omega(g) = \operatorname{ord}(g \cdot f)$, $s(g) = \omega(g) - |d(g)|$, $S(g) = \operatorname{supp} g$ and $\kappa = |S| - 1$. A *generator* is a primitive g with $s(g) \geq 0$.

Lemma 4.2 (primitive parts). *Every nonzero $a \in K[z]^{N+1}$ is $a = h\,g$ with $h = \gcd(a_0, \dots, a_N)$ and g primitive, unique up to a scalar. Two primitive vectors are $K(z)$-dependent if and only if they are proportional over K.*

Proof. If $pg = qg'$ with p, q coprime polynomials and g, g' primitive, then p divides every component of g', so p is constant, and likewise q. □

Since $f_0(0) \neq 0$, the vector e_0 has $\omega = 0$ and $s = -1$. So a generator with $|S| = 1$ is e_i with $i \geq 1$ and $\operatorname{ord} f_i \geq 1$.

Theorem 4.3 (union of elementary windows). *For a generator g put $W(g) = \{n : \exists h \neq 0,\ hg \in K(n)\}$. Then*

$$W(g) = \Big\{ n \in \mathbb{N}^{N+1} :\ n \geq d,\ \ |n-d| - \min_{i \in S} (n-d)_i \leq s \Big\} \qquad (d = d(g),\ s = s(g)),$$

and the zero set of the solid is $Z(\tau) = \bigcup_g W(g)$, over all generators.

Proof. By Lemma 2.2, $n \in Z$ if and only if some $a = hg \neq 0$ lies in $K(n)$. Now $hg \in K(n)$ if and only if $\deg h \leq \mu := \min_{i \in S}(n_i - d_i)$ (the degree conditions off S are void) and $\operatorname{ord} h \geq |n| - \omega$. Such an h exists if and only if $\mu \geq 0$ and $\mu \geq |n| - \omega$; take $h = z^{\max(0, |n|-\omega)}$. Since $|n| - \omega = |n-d| - s$, this is the displayed set. □

$W(g)$ is the union of the elementary blocks of the multiples $z^t g$. In the language of [BL 1992] it is the *combined singular block* of g, written here in all $N+1$ dimensions. The multiples hg with $h(0) \neq 0$ give nothing new, which is why non-primitive vectors can be discarded.

Theorem 4.4 (shape: the breathing simplex). *Let $\kappa \geq 1$ and $P_S(s) = \{x \in \mathbb{R}^{N+1}_{\geq 0} : |x| - \mathfrak{m}_S(x) \leq s\}$, so that $W(g) = d + (P_S(s) \cap \mathbb{Z}^{N+1})$. Then:*

1. *(H- and V-descriptions.) $P_S(s) = \{x \geq 0 :\ \sum_{j \neq i} x_j \leq s \ \ \forall i \in S\} = \operatorname{conv}\{0,\ se_0, \dots, se_N,\ \frac{s}{\kappa}\mathbf{1}_S\}$.*
2. *(Bipyramid.) $P_S(s) \cap \{|x| \leq s\} = \Delta_s$, and $P_S(s) \cap \{|x| \geq s\}$ is the pyramid over the facet $F_s = \{x \geq 0, |x| = s\}$ with apex $\frac{s}{\kappa}\mathbf{1}_S$. The window is two simplices glued along the order-s simplex.*

3. *(Order slices: the breathing simplex.) For* $0 \le r \le s(\kappa+1)/\kappa$,

$$P_S(s) \cap \{|x| = r\} = (r-s)_+ \mathbf{1}_S + L(r)\,\Delta^N, \qquad L(r) = \begin{cases} r, & r \le s, \\ (\kappa+1)s - \kappa r, & r \ge s. \end{cases}$$

Each slice is a simplex homothetic to the ambient order simplex and has the same orientation. It grows at rate 1 *until it fills the order-*s *simplex, then contracts at rate* κ *toward the barycentre of the face* S.

4. *(Gauge form.)* $P_S(s) = \{x \ge 0 : \ |x^\natural| + \kappa\,\mathfrak{m}_S(x) \le s\} = \bigcup_{0 \le t \le s/\kappa} (t\mathbf{1}_S + \Delta_{s-\kappa t})$. *The window is the corner simplex carried along the gauge orbit* $t\mathbf{1}_S$ *(multiplication by* z^t*), shrinking by* κ *per unit of massaction.*
5. *(Size.)* $\operatorname{vol} P_S(s) = (1+\frac{1}{\kappa})\frac{s^{N+1}}{(N+1)!}$ *and* $\#W(g) = \sum_{r=0}^{\lfloor s(\kappa+1)/\kappa \rfloor} \binom{L(r)+N}{N}$.

For $\kappa = 0$, *i.e.* $g = e_i$, *the window is the infinite prism* $\{x \ge 0 : \sum_{j \ne i} x_j \le s\}$, *the analogue of the infinite windows that leading zeros produce in a number wall.*

Proof. (1) $|x| - \min_{i \in S} x_i = \max_{i \in S} \sum_{j \ne i} x_j$, which gives the H-description. The function is convex ($-\min$ is convex) and the listed points satisfy it, so the hull lies in P. Conversely, let $x \in P$, $\mu = \mathfrak{m}_S(x)$ and $y = x - \mu\mathbf{1}_S \ge 0$. Then $|y| = |x| - (\kappa+1)\mu \le s - \kappa\mu$, so $\mu \le s/\kappa$. With $\lambda = \kappa\mu/s \in [0,1]$ we get $x = \lambda \cdot \frac{s}{\kappa}\mathbf{1}_S + y$ and $y \in (1-\lambda)\Delta_s$, so x lies in the hull. (2) If $|x| \le s$, the inequality holds automatically. If $|x| \ge s$, then $|x| \le s + \mu \le s + s/\kappa$. Set $\lambda = \kappa(|x| - s)/s \in [0,1]$ and $w = (x - \lambda\frac{s}{\kappa}\mathbf{1}_S)/(1-\lambda)$. Then $|w| = s$, and $w \ge 0$ because $\lambda s/\kappa = |x| - s \le \mu$. (3) On $|x| = r > s$ the inequality reads $\mu \ge r - s$, i.e. $x - (r-s)\mathbf{1}_S \ge 0$, a vector of total $r - (\kappa+1)(r-s)$. (4) This rewrites (1) using $|x| = |x^\natural| + (\kappa+1)\mathfrak{m}_S$. (5) The volume is $\operatorname{vol}\Delta_s$ plus the pyramid: $\frac{1}{N+1} \cdot \frac{\sqrt{N+1}\,s^N}{N!} \cdot \frac{s/\kappa}{\sqrt{N+1}}$. The count sums the lattice points of the slices. □

Remark (why the classical window is a square). For $N = 1$ and $S = \{0,1\}$ the H-description is $x_0 \le s$, $x_1 \le s$, the square $[0,s]^2$. Its antidiagonal slices have lengths $1, 2, \dots, s+1, \dots, 1$: growth at rate 1, contraction at rate $\kappa = 1$. In every dimension $N \ge 2$ a full-support window grows at rate 1 and contracts at rate N, so its upper pyramid holds only a fraction $1/N$ of the volume of the lower simplex. *The square's symmetry is the coincidence* $1 = \kappa$ *that happens only at* $N = 1$. Seen from the gauge, the shadow of a window along $\mathbf{1}_S$ is the order-s simplex, and over a class of standard-form size ℓ the window occupies a massaction interval of length $(s-\ell)/\kappa$. Certified on planted windows for $N = 1, 2, 3$ with full and partial supports, including every order slice (W5).

Theorem 4.5 (depth tent). *Let* $n \in W(g)$, $x = n - d$, *and suppose every nonzero element of* $K(n)$ *has primitive part* g *(the kernel is cyclic). Then*

$$\delta(n) = 1 + \min\left(\mathfrak{m}_S(x),\ s - |x^\natural| - \kappa\,\mathfrak{m}_S(x)\right) = 1 + \min\left(\mathfrak{m}_S(x),\ s - |x| + \mathfrak{m}_S(x)\right).$$

The maximum depth is $1 + \lfloor s/(\kappa+1) \rfloor$, *attained on the gauge axis* $x = t\mathbf{1}_S$, $t = \lfloor s/(\kappa+1) \rfloor$.

Proof. $K(n) = \{hg : \deg h \le \mu,\ \operatorname{ord} h \ge (|n| - \omega)_+\}$, which has dimension $\mu - (|n| - \omega)_+ + 1$. □

For $N = 1$ this is $1 + \min(x_0, x_1, s - x_0, s - x_1)$: one plus the distance to the edge of the square. Certified at 4886 single-window zeros (W4) and at every zero for $N = 1$ (W6).

Proposition 4.6 (general deficiency). *Let* $M_\sigma = \{a \in K[z]^{N+1} : \operatorname{ord}(a \cdot f) \ge \sigma\}$, *a free module of rank* $N+1$ *with* $\dim_K K[z]^{N+1}/M_\sigma = \sigma$. *Let* $b^{(0)}, \dots, b^{(N)}$ *be a basis of* $M_{|n|}$ *that is row-reduced for the shift* $-n$, *with shifted degrees* $e_j(n) = \max_i(\deg b_i^{(j)} - n_i)$. *Then* $\sum_j e_j(n) = 0$ *and*

$$\delta(n) = \sum_j \max(0, -e_j(n)) = \tfrac{1}{2}\|e(n)\|_1.$$

In particular $\tau(n) \ne 0$ *if and only if the reduced basis is balanced,* $e(n) = 0$.

Proof. The map $a \mapsto a \cdot f \bmod z^\sigma$ is onto $K[z]/(z^\sigma)$ because $f_0(0) \neq 0$, so the quotient has dimension $\sigma = \deg\det B$. For a shifted-reduced basis, $\deg\det B = \sum_j e_j + |n|$, and the predictable-degree property $\deg_n(\sum p_j b^{(j)}) = \max_j(\deg p_j + e_j)$ holds [Kailath 1980, Ch. 6; BL 1992, Thm. 3.1]. Then $K(n) = \{\sum p_j b^{(j)} : \deg p_j \le -1 - e_j\}$. □

The number of negative e_j, which is the $K(z)$-rank of $K(n)$, is at most N, because the e_j sum to zero. The next section extends this bound from a point to a cell.

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5. Separation becomes a rank bound

Theorem 5.1 (local rank bound). *Let $n^{(0)}, \dots, n^{(N)} \in \mathbb{N}^{N+1}$ satisfy*

$$\max_{\pi \in S_{N+1}} \sum_j n^{(j)}_{\pi(j)} \;-\; \min_j |n^{(j)}| \;\le\; N. \tag{5.1}$$

Then any $a^{(j)} \in K(n^{(j)})$ are linearly dependent over $K(z)$.

Proof. Let A be the matrix with rows $a^{(j)}$. Replacing its column 0 by $Af = \sum_i f_i\,(\text{column } i)$ multiplies the determinant by f_0, so $f_0 \det A = \sum_j (a^{(j)} \cdot f)\, C_j$ with polynomial cofactors C_j. Hence $\operatorname{ord}\det A \ge \min_j |n^{(j)}|$, using $f_0(0) \neq 0$. On the other hand $\deg\det A \le \max_\pi \sum_j (n^{(j)}_{\pi(j)} - 1)$, which by (5.1) is $< \min_j |n^{(j)}|$. So $\det A = 0$. □

Corollary 5.2 (unit cubes). *For every n, the kernels of the 2^{N+1} points of the unit cube $n + \{0,1\}^{N+1}$ together span a $K(z)$-space of dimension at most N. At most N independent generators have windows meeting any unit cube.*

Proof. For $v_j \in \{0,1\}^{N+1}$, $\max_\pi \sum_j v_{j\pi(j)} \le N+1$, with equality only if every v_j is nonzero, and then $\min_j |v_j| \ge 1$. So (5.1) holds. Distinct primitive vectors are $K(z)$-independent, and $n \in W(g)$ puts $z^t g$ in $K(n)$. □

Corollary 5.3 (the classical window theorem, $N = 1$). *In the number wall of $f = \sum c_k z^k$ (Example 2.3):*

1. *every zero lies in exactly one window;*
2. *the windows are the squares $d + [0,s]^2$ (from g with $S = \{0,1\}$), together with, when $\operatorname{ord} f = \omega \ge 1$, the infinite window $\{n_0 \le \omega - 1,\ n_1 \ge 1\}$ (from $g = e_1$);*
3. *no unit square meets two windows, so windows are separated in all eight directions and each has a closed frame of nonzero entries, corners included;*
4. *the depth formula $\delta = 1 + \min(x_0, x_1, s - x_0, s - x_1)$ holds at every zero.*

Proof. Corollary 5.2 with $N = 1$ gives (1) and (3) and makes every kernel cyclic. Theorems 4.4 and 4.5 give (2) and (4). □

This is the substance of the Padé block theorem and Lunnon's window theorem for one-sided walls. Here it falls out of the determinant inequality (5.1), without the cross rule. Certified (W6): 2940 unit squares, no violation.

Proposition 5.4 (touching and overlap, $N \ge 2$). *For $N = 2$, $f_0 = 1$, over $\mathbb{Q}$:*

1. *(Touching.) Take $f_1 = e^{z^2}$ and $f_2 = e^{z^2+z^3}$. The generator $(1,-1,0)$ has $d = (1,1,0)$ and $s = 0$, and its window is the point $(1,1,0)$. The generator $(0,1,-1)$ has $d = (0,1,1)$ and $s = 1$, and its window is the square pyramid $\{(0,1,1),(1,1,1),(0,2,1),(0,1,2),(0,2,2)\}$. The zeros $(1,1,0)$ and $(1,1,1)$ are adjacent along e_2 and lie in distinct windows.*

2. *(Overlap.) Take* $f_1 = e^{z^3}$ *and* $f_2 = e^{z^3+z^4}$. *Then* $\delta(1,1,1) = 2$, *since* $K(1,1,1)$ *consists of the constant vectors with zero sum. The point lies in the windows of* $(1,-1,0)$, $(1,0,-1)$ *and* $(0,1,-1)$, *each of which predicts depth* 1.

Over $\mathbb{F}_5$ *with random series, zeros of distinct windows are adjacent in all 26 king directions.* (Certified, W8.)

So for $N \geq 2$ both halves of the classical separation fail: windows touch, and windows overlap. Corollary 5.2 is what survives, and it is sharp: rank N is attained, even at a single point (part 2). The non-intersection failure itself was observed by [BL 1992] in two-dimensional cuts. The bound N is the new quantitative statement.

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6. The frame: what does not generalize

For $N = 1$ the frame theorem says that just outside each side of a window, the values form a geometric sequence (Lunnon). Its natural N-dimensional form would say that on the layer just outside each facet of the bipyramid, τ is geometric in every in-layer direction.

Observation 6.1 (the geometric facet law fails for $N = 2$). We planted windows over $\mathbb{F}_{101}$: $d = (3,3,3)$ with $s = 4$, and $d = (2,3,4)$ with $s = 3$. On the lower facet layers $\{n_i = d_i - 1,\ n + e_i \in W\}$, the test $\tau(n)\tau(n+2u) = \tau(n+u)^2$ fails in 34 of 36 and in all 18 of 18 in-layer coordinate instances. For the planted $N = 1$ window all instances hold (W9).

The mechanism is visible in Theorem 3.2. For $N = 1$, the cross rule at a frame point with a zero neighbour reduces to a two-term relation along a line, which forces geometric sequences. For $N \geq 2$, a zero vertex of a Hirota octahedron gives a two-term relation $\tau_j\tau_{ik} = \tau_k\tau_{ij}$ between points on different layers. These relations do not close into a law within one layer. (Exploratory checks, not part of the certificate, also found no two-dimensional cross rule on the coordinate planes through a window.)

Problem 6.2 (the frame for $N \geq 2$). Find the law governing τ on the boundary of an elementary window. The natural coordinates are probably the shifted degrees $e(n)$ of Proposition 4.6, not ratios of τ: at the frame the reduced basis is balanced but sits next to one with $\|e\|_1 = 2$. A solution would give the A_N number-wall algorithm, a Hirota–Miwa recursion that steps across windows as Lunnon's frame rules do in two dimensions. The look-ahead algorithms of Beckermann–Labahn and of Van Barel–Bultheel already do this numerically for Hermite–Padé forms.

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7. The window in one gauge

Collected in the language of Paper 01e:

- **The grid.** Each order slice of the solid is the finite simplex grid Σ_σ in barycentric coordinates. The solid is a stack of such grids, and Hirota–Miwa couples three consecutive layers through octahedra.
- **The gauge.** Multiplication by z moves a vector's degree data along $\mathbf{1}_S$. This is the gauge direction of the simplicial frame restricted to the face S, and it is not a symmetry: it raises the order by 1 while raising $|d|$ by $|S|$.
- **The window.** In standard form and massaction the window is the ball $|x^\natural| + \kappa\,\mathfrak{m}_S(x) \leq s$: the standard-form part plus κ times the massaction. Its shadow along the gauge is the order-s simplex. Its slices are simplices of the ambient orientation, never the inverted ones.
- **The square.** Conway's square window is the $N = 1$ breathing simplex. The order axis is the antidiagonal, the massaction axis is the diagonal, and the two rates agree.

The *finite simplex grid with strict capacity* that motivates the series appears here in a precise form. An approximation problem with a degree budget n and an order demand $|n|$ either has a unique solution ($\tau \neq 0$)

or overflows into a window. The window's size s is the excess order $\omega - |d|$ of the irreducible solution, and its shape is fixed by how the overflow propagates along the gauge.

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8. Open problems

1. **The frame (Problem 6.2).**
2. **Statistics over $\mathbb{F}_q$.** For random series, the number of elementary windows with excess s per unit volume, as a function of q, N and the support size. In our random runs over small fields most windows have $s = 0$ (single points), but zeros are far more frequent than in the $N = 1$ wall because windows may touch.
3. **Automatic solids.** For $N = 1$, the walls of automatic sequences mod p are two-dimensional automatic tilings [Lunnon 2009; ANL 2021; Robertson–Soffer Aranov 2025]. Are the solids of automatic tuples $(1, f_1, \ldots, f_N)$ automatic $(N+1)$-dimensional tilings, and what are their window fractals?
4. **Simultaneous Littlewood.** The number wall decides the t-adic Littlewood question for a single Laurent series. The Hermite–Padé solid is the corresponding object for simultaneous and multiplicative approximation of N series. Formulating the higher-dimensional Littlewood-type problems in terms of window sizes is the next natural target.

Appendix A: Certificates

Deterministic, local, no GPU (`numpy`, `fractions`). Criteria were declared in the script header before execution.

`empirical_01h/cert_window_theorem.py` — **10/10** (run 3; runs 1–2 logs retained). Run 1 failed W6 because non-primitive kernel vectors were counted as distinct generators; reducing to primitive parts (Definition 4.1) fixed it, and nothing else changed.

- **W1:** Hirota–Miwa exactly over $\mathbb{Q}$, $N = 2, 3$, 560 checks.
- **W2:** Hirota–Miwa over $\mathbb{F}_p$, $p \in \{2, 3, 5, 7\}$, 19476 checks.
- **W3:** $Z = \bigcup_g W(g)$ as sets, 36 solids ($N = 1, 2, 3$).
- **W4:** depth formula at 4886 single-window zeros.
- **W5:** planted windows: containment, lattice count, and every order slice equal to the predicted simplex, for $N = 1, 2, 3$ and full and partial supports.
- **W6:** $N = 1$: no unit square meets two windows (2940 squares); depth formula at all 1033 zeros.
- **W7:** unit-cube rank bound, $N = 2, 3$ (1014 cubes); rank 2 attained for $N = 2$.
- **W8:** Proposition 5.4 exactly over $\mathbb{Q}$; all 26 adjacency directions over $\mathbb{F}_5$.
- **W9:** Observation 6.1.
- **W10:** Proposition 3.3 exactly over $\mathbb{Q}$.

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